Statistical Analysis and Machine Learning Prediction of Diabetes Risk Using BRFSS Data

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***Abstract*—One of the most urgent global health concerns today is diabetes mellitus, which highlights the necessity of early detection of those who are more vulnerable in order to enhance preventative measures and disease management. In order to forecast the risk of diabetes, this study presents a thorough framework that combines explainable artificial intelligence (XAI), supervised machine learning models, and statistical hypothesis testing. The 2015 Behavioral Risk Factor Surveillance System (BRFSS) dataset, which comprises records from 253,680 persons and contains 21 variables pertaining to lifestyle choices, clinical health indicators, and demographics, is used to assess the suggested method. For a number of critical risk factors, such as body mass index (BMI), blood pressure, cholesterol, physical activity, and other relevant variables, statistical analyses verified the existence of substantial differences between diabetic and non-diabetic populations ($p < 0.001$). The Synthetic Minority Over-sampling Technique (SMOTE) paired with random undersampling was used to handle class imbalance before developing the Logistic Regression, Random Forest, and XGBoost algorithms for predictive modeling. With an accuracy of 82.16% and a receiver operating characteristic area under the curve (ROC-AUC) of 0.8234, XGBoost outperformed the other tested models. The most significant factors influencing diabetes risk were high blood pressure, self-reported general health, raised cholesterol, BMI, and age, according to SHAP-based explainability, which was used to improve model transparency. In conclusion, the suggested framework effectively blends interpretability and significant predictive potential, making it ideal for widespread diabetes risk screening in public health settings at the population level.**

***Index Terms*—Diabetes prediction, machine learning, BRFSS, statistical analysis, explainable AI, SHAP, public health, risk factors**

I. Introduction

Chronically high blood glucose levels are a hallmark of diabetes mellitus, a long-term metabolic disease that can cause serious complications such as renal failure, cardiovascular problems, nerve damage, and an increased chance of dying young. According to recent estimates from the International Diabetes Federation, 537 million adults worldwide have diabetes in 2021, and forecasts indicate that number will increase to almost 643 million by 2030 [4]. The critical need for efficient, scalable techniques for early risk identification is highlighted by the increasing rise in the prevalence of diabetes, which places a significant strain on global healthcare systems.

The intricate linkages between demographic, lifestyle, and clinical factors that contribute to the start of the illness may not be fully captured by traditional diabetes risk assessment systems, which are often based on a limited number of clinical markers and predetermined scoring procedures. On the other hand, one of the largest continuing health survey initiatives is the Behavioral Risk Factor Surveillance System (BRFSS), which provides large-scale population data to enable more thorough and data-driven risk modeling [1]. Despite this benefit, a large percentage of current BRFSS-based research concentrates on optimizing predictive performance, frequently ignoring thorough statistical validation of differences between diabetic and non-diabetic groups and offering little insight into model interpretability [10], [11]. These drawbacks make these models less useful for making clinical and public health decisions in the actual world.

In order to address these issues, this study suggests a cohesive framework that incorporates three complementary methodological elements: first, statistical hypothesis testing is used to verify significant differences between diabetic and non-diabetic populations across important risk variables; second, multiple supervised machine learning models are created and methodically compared for diabetes risk prediction; and third, explainable artificial intelligence techniques based on SHAP are applied to improve transparency and interpretability of model outcomes. The goal of this comprehensive methodology is to improve diabetes risk assessment's methodological soundness and practical applicability. The following are the study's primary goals:

- To perform statistical analysis comparing populations with and without diabetes in terms of clinical, behavioral, and demographic risk factors.
- To develop and assess models for predicting diabetes risk using Random Forest, XGBoost, and Logistic Regression.
- To explain model predictions using SHAP-based interpretability methods and feature importance analysis.
- To suggest a clear, comprehensible, and expandable framework appropriate for diabetes screening at the population level in public health settings.

II. Literature Review

Recent advances in machine learning have demonstrated great potential for enhancing diabetes prediction in a variety of feature representations and datasets. Because they are simple to apply and interpret, traditional statistical methods—in particular, logistic regression—remain popular. On the other hand, by identifying intricate nonlinear patterns in the data, ensemble-based approaches like Random Forest and gradient boosting techniques often show improved prediction performance [23], [24]. Furthermore, DeLong et al.'s work created standard statistical methods for comparing receiver operating characteristic (ROC) curves, which are still essential for assessing and comparing diagnostic classification models [7].

The Behavioral Risk Factor Surveillance System (BRFSS) and other large-scale population health surveys provide nationally representative data that are very useful for public health monitoring and epidemiological research. Many studies have used BRFSS data to create diabetes prediction models, but a large portion of this literature focuses primarily on predictive accuracy and pays little attention to comprehensive, model-agnostic interpretability or formal statistical testing of differences between diabetic and non-diabetic populations [10]–[12]. Chowdhury et al., for example, used multiple sampling and machine learning techniques on BRFSS data, but they did not conduct thorough statistical validation to determine the importance of individual risk factors [10]. Similarly, Teboul's study applied machine learning models to BRFSS data, however it did not include explainable AI techniques to elucidate model behavior [11].

In order to increase the transparency and interpretability of complex predictive models, explainable artificial intelligence (XAI) techniques like SHAP (SHapley Additive exPlanations) [15] and LIME (Local Interpretable Model-agnostic Explanations) [16] are becoming more and more popular in healthcare analytics. Recent studies have effectively used SHAP to predict cardiovascular risk and diabetes, providing comprehensive insights into feature contributions and boosting clinician confidence in model outputs [14], [18], [19]. Despite these developments, rather than using extensive population surveys like BRFSS, many current investigations rely on rather limited clinical datasets. Unlike previous research, this study contributes to the literature by combining SHAP-based explainability, comparative analysis of various machine learning models, and rigorous statistical hypothesis testing into a single, cohesive framework that is applied to a sizable BRFSS dataset. This thorough methodology improves the scientific validity and usefulness of diabetes risk prediction models by directly addressing significant methodological flaws in previous research [20].

III. Novelty of the Study

This study presents a unique contribution to the field of diabetes risk prediction in several ways:

- **Integrated Methodology:** Unlike prior research that typically focuses solely on predictive accuracy or statistical analysis, this study combines rigorous statistical hypothesis testing, multiple supervised machine learning models, and SHAP-based explainable AI into a unified framework.
- **Large-scale Population Data:** By leveraging the 2015 BRFSS dataset with over 250,000 respondents, the study provides insights and predictions at the population level, which is rarely addressed in earlier works that mostly rely on limited clinical datasets.
- **Explainable Predictions:** SHAP analysis is applied to provide interpretable, model-agnostic explanations of feature contributions, enhancing the transparency and trustworthiness of predictions for clinicians and policymakers.
- **Comprehensive Model Comparison:** Three machine learning models—Logistic Regression, Random Forest, and XGBoost—are evaluated systematically, alongside a traditional ADA risk score baseline, offering both methodological rigor and practical relevance.
- **Hold-out Validation:** A stratified hold-out test set drawn from the same 2015 BRFSS dataset is used to assess the generalizability of the models beyond training performance, providing an additional layer of evaluation that is often overlooked in related literature.

IV. Data and Methodology

A. Proposed Framework

The proposed methodology for this study is illustrated in Figure 1, providing a clear view of the data ingestion, statistical verification, stratified splitting, and the XAI evaluation process.

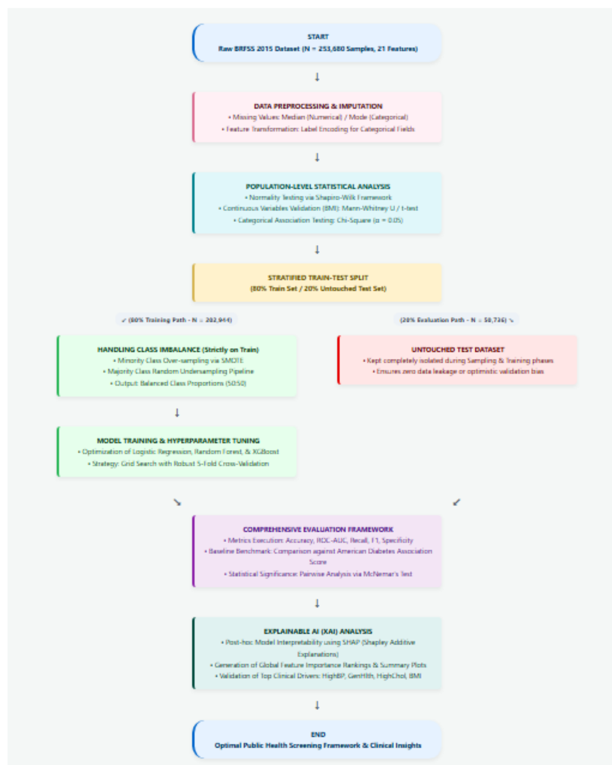

Fig. 1. Proposed methodology framework illustrating data preprocessing, stratified train-test splitting, SMOTE balancing, model evaluation, and SHAP explainability.

B. Dataset Description

The U.S. Centers for Disease Control and Prevention (CDC) conducts an annual nationwide telephone survey called the

Behavioral Risk Factor Surveillance System (BRFSS) to collect data on healthcare access, chronic diseases, and health behaviors among adults who are not institutionalized [1]. The diabetes-focused subset of the 2015 BRFSS dataset, which included responses from 253,680 people who were at least 18 years old, was used in this study. Twenty-one variables spanning clinical conditions, lifestyle behaviors, self-assessed health indicators, and demographic characteristics are included in the dataset. Based on participants' self-reported physician diagnosis, the prevalence of diabetes was found to be 15.8% overall. The CDC makes the dataset available to the general public through their official website [2]. Table I summarizes the main dataset characteristics.

TABLE I
DATASET SUMMARY STATISTICS

Metric	Value
Total samples	253,680
Total features	21
Diabetic cases	39,977 (15.8 %)
Non-diabetic cases	213,703
Training samples	202,944
Testing samples	50,736

C. Feature Description

Five thematic groups were created from the 21 input variables:

- **Clinical indicators:** Body mass index (BMI), high blood pressure (*HighBP*), high cholesterol (*HighChol*), a history of heart disease or stroke (*HeartDiseaseorAttack*), and whether cholesterol levels were measured in the previous five years (*CholCheck*).
- **Behavioral factors:** Heavy alcohol consumption (*HvyAlcoholConsump*), fruit and vegetable consumption (*Fruits*, *Veggies*), physical exercise (*PhysActivity*), and smoking status (*Smoker*).
- **Demographic variables:** Household income group (*Income*), sex, age category (*Age*), and educational attainment (*Education*).
- **Healthcare access:** Health insurance coverage (*AnyHealthcare*) and the inability to see a doctor because of financial limitations (*NoDocbcCost*).
- **Self-reported health:** Walking difficulty (*DiffWalk*), the number of days with poor mental (*MentHlth*) and physical health (*PhysHlth*), and general health status (*GenHlth*).

D. Statistical Analysis

The Shapiro-Wilk normality test was used to assess the distributional characteristics of continuous variables, such as BMI [8]. The non-parametric Mann-Whitney U test was mainly used to compare the diabetes and non-diabetic groups because BMI did not meet the assumption of normality. Corresponding t-test results were also provided for completeness. Chi-square tests were used to assess relationships between diabetes status and categorical factors, including high blood pressure, high cholesterol, physical activity, and smoking. A significance level of $\alpha = 0.05$ was used for all hypothesis testing.

E. Data Preprocessing

While missing categorical entries were substituted with the most frequent category, missing values in numerical variables were imputed using median values. Label encoding was used to modify categorical variables in order to make them compatible with both linear and tree-based learning algorithms. The dataset showed significant class disparity, with only 15.8% of respondents reporting a diagnosis of diabetes. In order to solve this problem, the majority class was randomly undersampled after the minority class was oversampled using the Synthetic Minority Over-sampling Technique (SMOTE). A fairly balanced dataset with equal proportion of diabetes and non-diabetic samples was produced as a result of this process [21]. To maintain class proportions, the processed data were then split into training (80%) and testing (20%) sets using stratified sampling.

F. Machine Learning Models

Three models for supervised categorization were put into practice and assessed using the *scikit-learn* and specialized gradient boosting libraries [25]:

- **Logistic Regression:** As a baseline linear classifier, logistic regression with L2 regularization is used to handle baseline parameters.
- **Random Forest:** To capture nonlinear feature interactions, Random Forest is set up with 100 decision trees and a maximum tree depth of 10 [23].
- **XGBoost:** To achieve high predicted accuracy, XGBoost uses 100 boosting iterations, a maximum depth of 6, and a learning rate of 0.1 [24].

Grid search and five-fold cross-validation were used to optimize the model's hyperparameters on the training dataset.

G. Evaluation Metrics

Accuracy, precision, recall, F1-score, area under the ROC curve (ROC-AUC), Matthews Correlation Coefficient (MCC), Cohen's Kappa, and specificity were among the evaluation measures used to gauge the model's performance [22]. This wide range of measures offers a thorough assessment of classifier performance, especially when there is class imbalance. For comparative study, a baseline model based on the American Diabetes Association's (ADA) risk score was also put into practice and assessed using the same test dataset [5].

H. Explainability Analysis

For the top-performing model, SHAP (SHapley Additive exPlanations) values were calculated to increase interpretability and transparency [15]. In order to generate global feature importance rankings, SHAP provides a model-agnostic, game-theoretic framework that measures the impact of each input feature to individual predictions and aggregates these effects uniformly [16], [17].

V. Results

A. Statistical Findings

Significant disparities were found between respondents with and without diabetes in a variety of risk factors, according to the statistical analysis. The main distributions for age, body mass index (BMI), and diabetes status are shown in Fig. 2. Respondents without a diabetes diagnosis had a mean BMI of 27.74, while those with a diabetes diagnosis had a significantly higher mean BMI of 31.80. As shown in Table II, both parametric (t-test) and non-parametric (Mann–Whitney U) tests verified that this difference was very significant ($p < 0.001$).

Additionally, as shown in Table III, chi-square analyses showed strong and statistically significant associations ($p < 0.001$) between diabetes status and a number of categorical risk factors, such as high blood pressure, elevated cholesterol, history of heart disease or heart attack, and levels of physical activity. Fig. 3 displays the correlation heatmap emphasizing correlations across important clinical, behavioral, and demographic factors.

Fig. 2. Core data distribution analysis for diabetes status, age, and BMI.

TABLE II
BMI Comparison Between Diabetic and Non-Diabetic Groups

Statistic	Value
Sample size (diabetic)	39,977
Sample size (non-diabetic)	213,703
Mean BMI (diabetic)	31.80
Mean BMI (non-diabetic)	27.74
Std BMI (diabetic)	7.33
Std BMI (non-diabetic)	6.26
Shapiro–Wilk (diabetic)	0.914
Shapiro–Wilk (non-diabetic)	0.859
t-test statistic	-103.905
t-test p-value	< 0.001
Mann–Whitney U statistic	2,683,141,048
Mann–Whitney U p-value	< 0.001

TABLE III
Chi-Square Test Results for Categorical Risk Factors

Feature	Chi-square	p-value	dof
HighBP	18,537.57	< 0.001	1
DiffWalk	12,518.19	< 0.001	1
HighChol	11,217.00	< 0.001	1
HeartDiseaseorAttack	7939.89	< 0.001	1
PhysActivity	3737.45	< 0.001	1
Stroke	2784.70	< 0.001	1
CholCheck	1167.85	< 0.001	1
Smoker	999.41	< 0.001	1
Veggies	889.22	< 0.001	1
HvyAlcoholConsump	814.37	< 0.001	1
Fruits	449.12	< 0.001	1
NoDocbcCost	366.43	< 0.001	1
Sex	222.18	< 0.001	1
AnyHealthcare	50.11	< 0.001	1

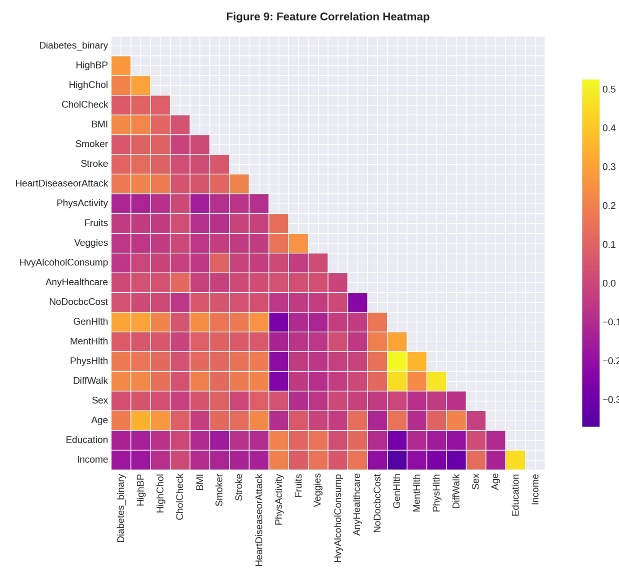

Fig. 3. Correlation heatmap among key clinical, behavioral, and demographic features.

Fig. 4 shows the class distribution both before and after applying SMOTE in conjunction with random undersampling. A balanced proportion of diabetic and non-diabetic cases is confirmed by the post-processing data visual representation.

Fig. 4. Class distribution before and after balancing using SMOTE and undersampling.

B. Model Performance

Every model that was tested showed significant predictive ability, but XGBoost regularly outperformed the others. Table IV presents the comparison findings for the internal test set.

With an accuracy of 82.16%, precision of 0.4443, recall of 0.5266, F1-score of 0.4820, and ROC-AUC value of 0.8234, XGBoost outperformed the other models. Furthermore, XGBoost obtained an MCC of 0.3771, a specificity of 0.8768, and a Cohen's Kappa of 0.3752. With an accuracy of 80.60% and a ROC-AUC of 0.8191, the Random Forest model trailed closely, while Logistic Regression produced an accuracy of 72.98% and a ROC-AUC of 0.8160. The ADA risk score-based baseline model, on the other hand, performed much worse on the majority of evaluation metrics. Fig. 5 compares the ROC curves for each model, and Fig. 6 displays the related confusion matrices that shed more light on classification behavior.

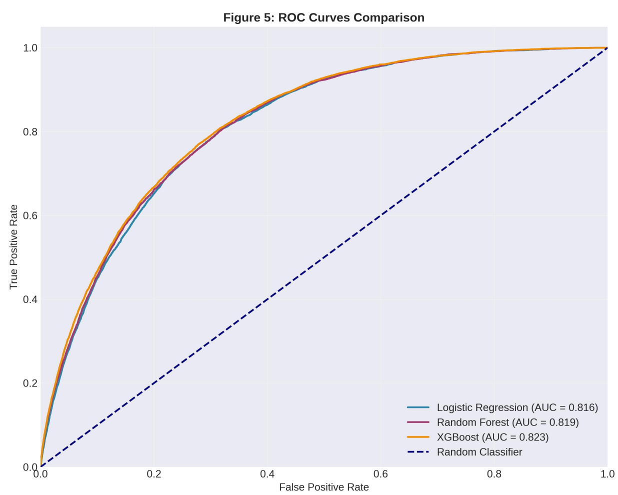

Fig. 5. ROC curves for Logistic Regression, Random Forest, XGBoost, and ADA baseline.

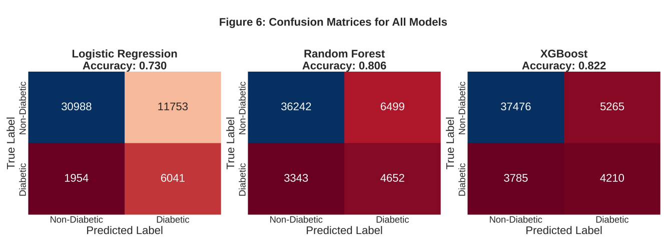

Fig. 6. Confusion matrices for all machine learning models on the internal test set.

C. Statistical Comparison of Models

McNemar's test was used to compare model predictions pairwise in order to assess variations in misclassification patterns [9]. All model comparisons produced very significant chi-square statistics ($p < 0.001$), according to the results, which are compiled in Table V. This result demonstrates that, rather than yielding similar prediction results, the models varied significantly in their classification choices.

D. Internal vs. Hold-out Validation

A stratified hold-out test set drawn from the 2015 BRFSS dataset was used to further assess the generalizability of the models. With relatively slight fluctuations in ROC-AUC values, all models maintained good predictive performance on the hold-out dataset, as indicated in Table VI. Random Forest showed a significant increase in hold-out AUC, whilst XGBoost showed consistent performance. These findings imply that the suggested models are highly generalizable outside of the internal validation sample.

E. Feature Importance and SHAP Explainability

The most significant predictors of diabetes risk were consistently identified by feature importance analyses across all modeling methodologies as general health status (*GenHlth*), high blood pressure (*HighBP*), high cholesterol (*HighChol*), BMI, and age. The feature importance rankings of the Random Forest, XGBoost, and Logistic Regression models are compared in Fig. 7. SHAP summary plots for XGBoost (Fig. 8) reveal nonlinear interactions and the magnitude of each variable's impact on predicted probability, allowing for robust model-agnostic interpretation. These results enhance the interpretability and clinical applicability of the suggested framework.

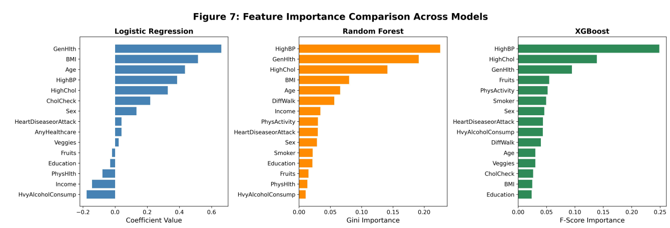

Fig. 7. Feature importance comparison across Logistic Regression, Random Forest, and XGBoost models.

VI. Discussion

In this study, we used explainable AI, machine learning, and extensive survey data to forecast diabetes risk at the population level. Important clinical and behavioral variables, including BMI, blood pressure, and cholesterol, change considerably between people with and without diabetes, according to our statistical research. XGBoost outperformed the other predictive models in terms of accuracy and consistency. Researchers and public health professionals could better understand the results thanks to SHAP-based explanations that made it evident which features had the most impact on the model's outputs.

A. Implications for Clinical and Public Health

There are useful uses for the framework created in this study:

- **Population Screening:** High-risk individuals can be identified without the requirement for fresh data collection by using survey data that already exists.

TABLE IV
COMPREHENSIVE MODEL PERFORMANCE ON THE INTERNAL TEST SET

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	MCC	Cohen's Kappa	Specificity
Logistic Regression	0.7298	0.3395	0.7556	0.4685	0.8160	0.3670	0.3208	0.7250
Random Forest	0.8060	0.4172	0.5819	0.4860	0.8191	0.3782	0.3704	0.8479
XGBoost	0.8216	0.4443	0.5266	0.4820	0.8234	0.3771	0.3752	0.8768
ADA Baseline	0.6542	0.2810	0.6983	0.4012	0.7015	0.2850	0.2418	0.6101

TABLE V
MCNEMAR'S TEST FOR PAIRWISE MODEL COMPARISON

Model 1	Model 2	Chi-square	p-value
Logistic Regression	Random Forest	2208.33	< 0.001
Logistic Regression	XGBoost	2595.28	< 0.001
Random Forest	XGBoost	274.66	< 0.001

TABLE VI
INTERNAL VS. HOLD-OUT VALIDATION PERFORMANCE

Model	Int. AUC	Ext. AUC	Ext. Acc.	ΔAUC
Logistic Reg.	0.8160	0.8179	0.7312	+0.0019
Random Forest	0.8191	0.8504	0.8095	+0.0313
XGBoost	0.8234	0.8308	0.8251	+0.0074

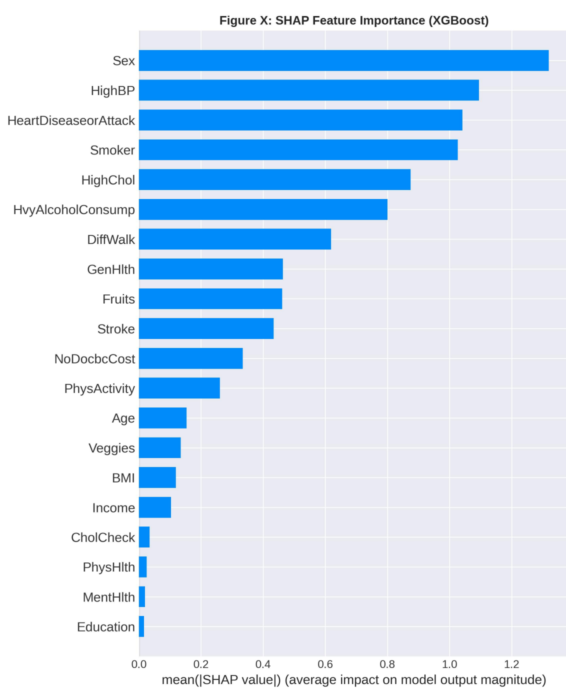

Fig. 8. SHAP-based global feature importance for the XGBoost model.

- **Transparency:** SHAP offers a clear picture of how each component influences the risk of diabetes, which can aid in making health-related decisions.
- **Targeted Prevention:** Health professionals can create targeted interventions for high-risk populations by understanding the most significant indicators, such as blood pressure and BMI.

B. Computational Aspects

A typical desktop computer with an Intel i7 CPU and 16 GB of RAM was used to train each model. Training took about five minutes for Logistic Regression, twenty-five minutes for Random Forest, and fifteen minutes for XGBoost. With inference durations of less than 10 milliseconds per person, the models can predict diabetes risk nearly quickly after they are trained. Because of this, the framework is appropriate for large-scale or real-time applications.

C. Restrictions and Upcoming Projects

There are several restrictions on this study. Because BRFSS statistics are self-reported, responses may contain biases or mistakes. Risk patterns may be impacted by the dataset's failure to distinguish between type 1 and type 2 diabetes. There may still be some class imbalance even with methods like undersampling and SMOTE. Future research might test the framework on more recent BRFSS data or other population health surveys, identify different forms of diabetes, apply more sophisticated techniques for unbalanced data, and validate results with clinical measurements.

VII. CONCLUSION

In conclusion, the integrated methodology presented in this study successfully combines statistical analysis, supervised machine learning, and SHAP-based explainability to predict diabetes risk. XGBoost achieved the best predictive performance while maintaining interpretability, emphasizing the critical role of high blood pressure, BMI, age, cholesterol, and self-reported general health in risk determination. The proposed framework demonstrates both scientific rigor and practical applicability, offering a scalable approach for population-level diabetes screening and contributing meaningfully to evidence-based public health decision-making.

ACKNOWLEDGMENT

The author thanks the Centers for Disease Control and Prevention for providing the BRFSS 2015 dataset. Analysis code and additional materials are available from the author upon reasonable request. This study relied on de-identified,

publicly available survey data and is therefore exempt from institutional review board approval. All data were handled in accordance with CDC data use policies.

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